Target validation paths — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

The whole plan hinges on three pending gates, none of them resolved. High-resolution HLA class I typing plus HA-1/HA-2 genotyping decide whether donor TCR-T (TSC-100/TSC-101 via NCT05473910) is reachable at all; a variant-level relapse NGS panel decides whether the Y220C-selective reactivator rezatapopt (via PYNNACLE, NCT04585750) is in play; and STR chimerism on sorted CD34+ blasts decides whether the relapse is even recipient-derived and therefore targetable by minor-antigen T cells. If HLA-A*02:01 comes back absent, the HA-1/HA-2 and WT1 TCR-T levers both close, and the next conversation about standard salvage care belongs to the treating team rather than to this report.

HLA class I typing and HA-1/HA-2 minor antigens

Essential, and the first thing to do. Retrieve the 2020 allele-level HLA-A typing (report A*02:01 and A*24:02) from the transplant or donor-registry chart, same-day if the record exists and 3-7 days to repeat if it cannot be found. A*02:01 is the master switch for the entire restricted axis. The same A*02:01 result now unlocks three A*02:01-restricted levers rather than two: alongside HA-1/HA-2 TCR-T and A*02:01 WT1 TCR-T, a positive type opens CBX-250, the off-the-shelf CG1/HLA-A*02:01 T-cell engager in CROSSCHECK-001 (NCT06994676), which needs documented A*02:01 alone and no HA-1/HA-2 mismatch. In parallel, genotype HA-1/HA-2 in both the patient and the HLA-identical sibling donor: TSC-100 (HA-1) and TSC-101 (HA-2) via NCT05473910 require a recipient-positive, donor-negative mismatch, and HLA-identical siblings can still differ here. This is a single-SNP PCR genotype, roughly 1-2 weeks, most reliably run through the ALLOHA sponsor central lab because standalone HA-1/HA-2 testing is not a routine catalog assay. Skipping the donor arm leaves eligibility indeterminate even when the patient genotype looks favorable.

TP53

Held aberrant pending confirmation, not resolved in either direction. A comprehensive relapse myeloid NGS panel with variant-level readout and copy-number detection (1-3 weeks) is exactly what the 2026 relapse panel lacked, given its 2% limit of detection and absence of copy-number calling; it names the specific amino-acid change and gates rezatapopt via NCT04585750, which needs a confirmed Y220C. Read it together with three high-priority orthogonal checks rather than trusting any one alone: TP53 (17p13) FISH for copy loss, p53 IHC (clone DO-7) for an aberrant overexpression or null pattern, and retrieval of the 2019 diagnostic report for the original variant identity and VAF. Do not down-call to wild-type on the strength of the limited relapse panel.

Second-transplant platform

STR chimerism on FACS-sorted CD34+ blasts (1-2 weeks) is the load-bearing test here. It confirms whether the relapse is recipient-derived evolution of the original clone or a donor-derived leukemia, a distinction the sex-matched sibling donor keeps XY-FISH from settling. Cell sorting is the step that matters, since a bulk-marrow result cannot resolve blast-clone origin, and recipient origin is a prerequisite for any HA-directed TCR-T. Alongside it, a full relapse metaphase karyotype characterizes the adverse MECOM rearrangement and KMT2A amplification for risk stratification and trial eligibility, and right-thigh contrast MRI plus whole-body FDG-PET/CT maps the extramedullary myeloid sarcoma for a directed-radiotherapy decision before or bridging to the transplant.

CD33

Quantitative CD33 antigen-density flow (antibodies-bound-per-cell, 3-5 days) upgrades the current qualitative-positive call into the metric that predicts gemtuzumab ozogamicin benefit. Request the density readout explicitly; a standard panel reports positivity, not the antigen density that drives ADC selection.

CD123

Quantitative CD123 antigen-density flow, batched on the same fresh draw to conserve marrow, guides CD123-directed selection (tagraxofusp, pivekimab sunirine) and dual CD33/CD123 cytoreduction ahead of any cellular or transplant consolidation.

Where to order these assays

Assay	Provider	Decision gated	Contact
High-resolution HLA class I typing (A02:01, A24:02)	**ARUP Laboratories (preferred) (HLA Class I High-Resolution Typing)**	**HA-1/HA-2 TCR-T (TSC-100/TSC-101 via NCT05473910) plus WT1- and PRAME-directed A02:01-restricted immunotherapy; A24:02 opens a WT1-siTCR alternative.**	**test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787**
High-resolution HLA class I typing (A02:01, A24:02)	Versiti Diagnostic Laboratories	HA-1/HA-2 TCR-T (TSC-100/TSC-101 via NCT05473910) plus WT1- and PRAME-directed A02:01-restricted immunotherapy; A24:02 opens a WT1-siTCR alternative.	test info · 638 N 18th St, Milwaukee, WI 53233
High-resolution HLA class I typing (A02:01, A24:02)	Labcorp	HA-1/HA-2 TCR-T (TSC-100/TSC-101 via NCT05473910) plus WT1- and PRAME-directed A02:01-restricted immunotherapy; A24:02 opens a WT1-siTCR alternative.	test info · 358 South Main St, Burlington, NC 27215 · 800-845-6167
High-resolution HLA class I typing (A02:01, A24:02)	Bloodworks Northwest HLA Laboratory	HA-1/HA-2 TCR-T (TSC-100/TSC-101 via NCT05473910) plus WT1- and PRAME-directed A02:01-restricted immunotherapy; A24:02 opens a WT1-siTCR alternative.	test info · 1551 Eastlake Ave E, Seattle, WA 98102
HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient + sibling donor)	**TScan Therapeutics (preferred) (ALLOHA HA-1 eligibility genotyping)**	**TSC-100 (HA-1) / TSC-101 (HA-2) donor-derived TCR-T via NCT05473910; requires recipient-positive / donor-negative mHAg mismatch.**	**test info · 830 Winter St, Waltham, MA 02451**

HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient + sibling donor)	Fred Hutchinson Cancer Center (Bleakley program)	TSC-100 (HA-1) / TSC-101 (HA-2) donor-derived TCR-T via NCT05473910; requires recipient-positive / donor-negative mHAg mismatch.	test info · 1100 Fairview Ave N, Seattle, WA 98109
HA-1 / HA-2 minor histocompatibility antigen genotyping (recipient + sibling donor)	Versiti Diagnostic Laboratories	TSC-100 (HA-1) / TSC-101 (HA-2) donor-derived TCR-T via NCT05473910; requires recipient-positive / donor-negative mHAg mismatch.	test info · 638 N 18th St, Milwaukee, WI 53233
Comprehensive relapse myeloid NGS (variant-level TP53 + CNV)	**Foundation Medicine (preferred) (FoundationOne Heme)**	**Rezatapopt (PC14586) via PYNNACLE (NCT04585750) if Y220C-positive; also re-tests FLT3/IDH1/IDH2 at relapse.**	**test info · 150 Second St, Cambridge, MA 02141 · 888-988-3639**
Comprehensive relapse myeloid NGS (variant-level TP53 + CNV)	Tempus (Tempus xT Heme)	Rezatapopt (PC14586) via PYNNACLE (NCT04585750) if Y220C-positive; also re-tests FLT3/IDH1/IDH2 at relapse.	test info · 600 West Chicago Ave, Suite 510, Chicago, IL 60654 · 800-739-4137
Comprehensive relapse myeloid NGS (variant-level TP53 + CNV)	NeoGenomics Laboratories (NeoTYPE Myeloid Disorders Profile)	Rezatapopt (PC14586) via PYNNACLE (NCT04585750) if Y220C-positive; also re-tests FLT3/IDH1/IDH2 at relapse.	test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907
Comprehensive relapse myeloid NGS (variant-level TP53 + CNV)	Mayo Clinic Laboratories (OncoHeme Next-Generation Sequencing)	Rezatapopt (PC14586) via PYNNACLE (NCT04585750) if Y220C-positive; also re-tests FLT3/IDH1/IDH2 at relapse.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
TP53 (17p13) FISH for deletion / copy loss	**NeoGenomics Laboratories (preferred) (TP53/17p FISH)**	**Confirms TP53-aberrant (17p loss / biallelic) call underpinning prognosis and rezatapopt candidacy workup.**	**test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907**
TP53 (17p13) FISH for deletion / copy loss	Mayo Clinic Laboratories	Confirms TP53-aberrant (17p loss / biallelic) call underpinning prognosis and rezatapopt candidacy workup.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
TP53 (17p13) FISH for deletion / copy loss	ARUP Laboratories	Confirms TP53-aberrant (17p loss / biallelic) call underpinning prognosis and rezatapopt candidacy workup.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787
p53 protein IHC on marrow biopsy (clone DO-7)	**NeoGenomics Laboratories (preferred) (p53 IHC)**	**Confirms TP53-aberrant target call at the protein level; does not establish Y220C (NGS required).**	**test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907**
p53 protein IHC on marrow biopsy (clone DO-7)	Mayo Clinic Laboratories	Confirms TP53-aberrant target call at the protein level; does not establish Y220C (NGS required).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
p53 protein IHC on marrow biopsy (clone DO-7)	ARUP Laboratories	Confirms TP53-aberrant target call at the protein level; does not establish Y220C (NGS required).	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787

STR chimerism on FACS-sorted CD34+ blasts	**Mayo Clinic Laboratories (preferred) (Chimerism, Cell-Sorted, STR)**	**Recipient-origin relapse is a prerequisite for HA-directed TCR-T and steers DLI / HCT2 strategy.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710**
STR chimerism on FACS-sorted CD34+ blasts	ARUP Laboratories	Recipient-origin relapse is a prerequisite for HA-directed TCR-T and steers DLI / HCT2 strategy.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787
STR chimerism on FACS-sorted CD34+ blasts	CareDx (AlloSeq HCT)	Recipient-origin relapse is a prerequisite for HA-directed TCR-T and steers DLI / HCT2 strategy.	test info · 8000 Marina Blvd, Suite 4600, Brisbane, CA 94005 · 415-287-2300
Quantitative CD33 antigen-density flow	**Hematologics, Inc. (preferred) (Difference-from-Normal flow with antigen quantitation)**	**Gemtuzumab ozogamicin (CD33 ADC) candidacy; informs dual CD33/CD123 targeting.**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · 206-223-4553**
Quantitative CD33 antigen-density flow	NeoGenomics Laboratories	Gemtuzumab ozogamicin (CD33 ADC) candidacy; informs dual CD33/CD123 targeting.	test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907
Quantitative CD33 antigen-density flow	Mayo Clinic Laboratories	Gemtuzumab ozogamicin (CD33 ADC) candidacy; informs dual CD33/CD123 targeting.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Quantitative CD123 antigen-density flow	**Hematologics, Inc. (preferred) (Difference-from-Normal flow with antigen quantitation)**	**Tagraxofusp / pivekimab sunirine (CD123-directed) selection and dual CD33/CD123 cytoreduction.**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · 206-223-4553**
Quantitative CD123 antigen-density flow	NeoGenomics Laboratories	Tagraxofusp / pivekimab sunirine (CD123-directed) selection and dual CD33/CD123 cytoreduction.	test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907
Quantitative CD123 antigen-density flow	Mayo Clinic Laboratories	Tagraxofusp / pivekimab sunirine (CD123-directed) selection and dual CD33/CD123 cytoreduction.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Full metaphase karyotype at relapse	**NeoGenomics Laboratories (preferred) (Chromosome Analysis, Bone Marrow)**	**AML risk stratification and trial-eligibility cytogenetics; supports clonal-evolution comparison.**	**test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907**
Full metaphase karyotype at relapse	Mayo Clinic Laboratories	AML risk stratification and trial-eligibility cytogenetics; supports clonal-evolution comparison.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
Full metaphase karyotype at relapse	ARUP Laboratories	AML risk stratification and trial-eligibility cytogenetics; supports clonal-evolution comparison.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787

The 2019-report retrieval and the extramedullary-disease imaging are not catalog assays: the first is a records or archived-DNA request to the 2019 originating laboratory named in that pathology report, and the second is performed by the treating institution's radiology and nuclear-medicine services.

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
High-resolution HLA class I typing (HLA-A allele level; report A02:01 and A24:02)	Resolves HLA-A02:01 status, the master switch for the HLA-restricted immunotherapy axis that motivated this case: HA-1/HA-2 TCR-T (TSC-100/TSC-101), WT1, and PRAME approaches are all A02:01-restricted. Parallel A24:02 typing preserves a WT1-siTCR option if A02:01 is negative. The 2020 allele-level result is only a placeholder in the chart, so retrieval or repeat high-resolution typing gates the entire pathway before any of these interventions can be scoped.	ARUP Laboratories (HLA Class I High-Resolution Typing) · test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 800-522-2787	Pull the archived 2020 high-resolution typing from the transplant/donor-registry file; if not retrievable, 5-10 mL EDTA (lavender-top) whole blood or a buccal swab
HA-1 / HA-2 minor histocompatibility antigen genotyping of recipient AND HLA-identical sibling donor	HA-directed TCR-T requires a specific mismatch, recipient HA-1(H)-positive and donor HA-1-negative, and HLA-identical siblings can still differ at HA-1/HA-2. Genotyping both the patient and the sibling donor establishes whether TSC-100 (HA-1) or TSC-101 (HA-2) eligibility exists; the approach is additionally gated on HLA-A*02:01. Skipping the donor arm leaves eligibility indeterminate even when the patient genotype is favorable.	TScan Therapeutics (ALLOHA trial central testing) (ALLOHA HA-1 eligibility genotyping) · test info · 830 Winter St, Waltham, MA 02451	5-10 mL EDTA (lavender-top) whole blood from the patient and a paired sample from the sibling donor (buccal swab acceptable for the SNP genotype)

Comprehensive relapse myeloid NGS panel with variant-level TP53 readout, low-VAF sensitivity, and copy-number detection	A relapse myeloid NGS panel with variant-level readout, adequate low-VAF sensitivity, and copy-number detection resolves the specific TP53 amino-acid change and reconciles the 2019-pathogenic versus 2026-negative discrepancy without down-calling on a 2%-LOD/no-CNV assay. A confirmed Y220C opens the Y220C-selective reactivator rezatapopt (PC14586); a non-Y220C aberration would not, and eprenetapopt is low-yield. The same panel re-interrogates FLT3, IDH1, and IDH2 at relapse given documented clonal evolution.	Foundation Medicine (FoundationOne Heme) · test info · 150 Second St, Cambridge, MA 02141 · 888-988-3639	Relapse bone marrow aspirate (EDTA/heparin) or peripheral blood at current high blast burden; retrieve archived 2019 fixed-cell pellet or extracted DNA for side-by-side comparison
TP53 (17p13) FISH for deletion / copy loss	TP53 (17p13) FISH detects allelic copy loss and 17p deletion that the 2026 relapse panel could not see because it lacked copy-number detection, which is central to reconciling the discrepancy without assuming reversion to wild-type. Biallelic (multi-hit) TP53 status refines both prognosis and the strength of the TP53-aberrant call. Omitting it risks mis-classifying a copy-loss-driven aberration as absent.	NeoGenomics Laboratories (TP53/17p FISH) · test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907	Bone marrow aspirate (fresh, sodium heparin) or an archival fixed-cell pellet / marrow smears from the relapse workup

p53 protein on marrow biopsy (clone DO-7)	Strong or aberrant (null-pattern) p53 expression by immunohistochemistry on the marrow biopsy is a protein-level correlate of TP53 mutation and multi-hit status in myeloid neoplasms. A positive pattern supports the TP53-aberrant working call independent of the NGS discrepancy, whereas a clean wild-type pattern would prompt scrutiny of the 2019 result. It does not establish the Y220C variant, so it complements rather than replaces variant-level NGS.	NeoGenomics Laboratories (p53 IHC) · test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907	FFPE bone marrow trephine core biopsy or clot section; archival material from the relapse marrow is acceptable
Retrieval of the 2019 diagnostic NGS report (TP53 variant identity + VAF)	Retrieving the original 2019 diagnostic report gives the exact TP53 variant identity and VAF that anchor the reconciliation; a founder mutation shared with the 2019 clone would argue the 2026 negative reflects subclonal loss or assay limitation rather than true reversion. Whether that variant is Y220C directly determines rezatapopt candidacy. Without the original identity the discrepancy cannot be adjudicated and the therapy handle stays undefined.	2019 originating laboratory named in the pathology report (records / archived-DNA request; not a catalog assay)	No new specimen; obtain the 2019 diagnostic NGS report and, if re-testing is needed, the archived 2019 fixed-cell pellet or extracted DNA from the originating laboratory

STR chimerism on FACS-sorted CD34+ blasts with side-by-side 2019-vs-2026 cytogenetic comparison	STR chimerism on FACS-sorted CD34+ blasts, read alongside a side-by-side 2019-versus-2026 cytogenetic comparison, confirms whether the relapse is recipient-derived evolution of the original clone or a donor-derived leukemia (the sex-matched sibling donor precludes XY-FISH). Recipient origin is a prerequisite for HA-directed TCR-T, which targets recipient-restricted minor histocompatibility antigens, and it shapes DLI and second-transplant strategy. A donor-derived leukemia would redirect the entire plan.	Mayo Clinic Laboratories (Chimerism, Cell-Sorted, STR) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710	Relapse bone marrow with CD34+ cell sorting, plus paired recipient (pre-transplant or buccal) DNA and donor reference DNA
Quantitative CD33 antigen-density flow cytometry (antibodies bound per cell)	Quantitative CD33 antigen density upgrades the current qualitative-positive flow to the metric that predicts benefit from the CD33 ADC gemtuzumab ozogamicin; higher blast CD33 expression correlates with greater gemtuzumab effect. Density also informs dual CD33/CD123 targeting. Qualitative positivity alone does not establish whether ADC-level expression is present.	Hematologics, Inc. (Difference-from-Normal flow with antigen quantitation) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · 206-223-4553	Fresh (viable) bone marrow aspirate or peripheral blood in EDTA/heparin, delivered within 24-48 h for antigen-density quantitation

Quantitative CD123 antigen-density flow cytometry (antibodies bound per cell)	Quantitative CD123 antigen density converts the qualitative-positive call into the expression metric that guides CD123-directed selection: tagraxofusp (CD123 immunotoxin) and pivekimab sunirine (CD123 ADC). Density supports agent choice and dual CD33/CD123 cytoreduction ahead of a cellular or transplant consolidation. A density readout de-risks committing to a CD123 agent on qualitative positivity alone.	Hematologics, Inc. (Difference-from-Normal flow with antigen quantitation) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · 206-223-4553	Fresh (viable) bone marrow aspirate or peripheral blood in EDTA/heparin, delivered within 24-48 h for antigen-density quantitation
Full metaphase karyotype (conventional cytogenetics) at relapse	A full metaphase karyotype at relapse completes the FISH-based 2026 picture and characterizes the adverse MECOM rearrangement and KMT2A amplification that shape prognosis and trial eligibility. Complete cytogenetics is required by most AML risk models and trial protocols and supports the side-by-side clonal-evolution comparison. FISH panels alone can miss balanced rearrangements and additional subclones.	NeoGenomics Laboratories (Chromosome Analysis, Bone Marrow) · test info · 12701 Commonwealth Drive, Suite 9, Fort Myers, FL 33913 · 866-776-5907	Fresh bone marrow aspirate in sodium heparin (green-top) for metaphase culture

Right-thigh contrast MRI plus whole-body FDG-PET/CT for extramedullary disease mapping	Contrast MRI of the right thigh plus whole-body FDG-PET/CT maps the extent of the extramedullary myeloid sarcoma, an adverse feature, and shows whether directed radiotherapy is warranted before or bridging to a second transplant. Extramedullary burden also affects conditioning choice (for example targeted radioimmunotherapy) and response assessment. Leaving it unmapped risks under-treating a sanctuary site that drives relapse.	Treating institution's radiology and nuclear-medicine services (not a reference laboratory)	No tissue required; contrast-enhanced MRI of the right thigh and whole-body FDG-PET/CT

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.